

THE STRICT LAYER OF GENERIC CELLS IN CUBICAL TYPE THEORY IS A WEDGE OF DE MORGAN SPHERES

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ABSTRACT. In cubical type theories with a De Morgan interval, the connection and reversal operations give types strictly more definitional structure than the identity types of Martin-Löf type theory: many groupoid laws that are only propositional in Book HoTT hold judgmentally. We determine exactly how much. Over a context of generic higher cells, the cubes definable by pure reparametrization — without Kan operations — form, modulo definitional equality and naturally in the cube dimension, a *wedge of De Morgan spheres*: one boundary-collapsed representable presheaf $\square_{\text{DM}}(-, [n])/\partial$ on the De Morgan cube category per cell. The action of the cube category is strictly functorial (identities and composition hold definitionally), it collapses precisely the morphisms that factor through a face, it is faithful elsewhere, and distinct cells never interact. In particular transposition of a 2-cell’s coordinates is not definitional — the exact definitional shadow of the Eckmann–Hilton argument being weak — and a square such as $q(i \wedge \neg i)(j)$, all four of whose edges are degenerate, is not definitionally constant: the non-Boolean nature of the free De Morgan algebra is visible in every dimension. All positive and negative instances are machine-checked in a self-contained proof-assistant kernel, and by the metatheory of a companion paper every statement is a decidable, assumption-free fact about the algorithmic equality of a minimal fragment.

1. INTRODUCTION

A basic discovery of homotopy type theory is that the identity types of Martin-Löf type theory equip every type with the structure of a weak ω -groupoid [4, 5]. The structure is *weak*: associativity, units, inverses and the exchange laws hold only up to higher cells, and the bookkeeping of those higher cells — the coherence problem — is notoriously difficult.

Cubical type theories [1] change the balance. Their interval $\mathbb{I}$ carries the operations of a De Morgan algebra ($\wedge, \vee, \neg$), and *reparametrizing* a cell along these operations — forming $\langle i \rangle p(i \wedge j)$ and its relatives, the *connections* — is a definitional operation: many laws that Book HoTT can only prove propositionally hold judgmentally. Practitioners know this well [7]; what has

2020 *Mathematics Subject Classification.* 18N45, 03B38, 18M99.

Key words and phrases. cubical type theory, cube categories, connections, De Morgan algebra, coherence, definitional equality, weak ω -groupoids.

been missing is a precise statement of *how much* of the ω -groupoid structure the definitional equality supplies, and where exactly the strict layer ends and the weak layer necessarily begins.

This paper answers that question for the layer generated by *generic cells*. Fix base types and points, and let q be a generic n -cell — a variable of the n -fold iterated loop type. The definable cubes over such a context, using only reparametrization (no Kan operations), carry an action of the *De Morgan cube category* $\square_{\text{DM}}$: objects $[m]$, morphisms $[m] \rightarrow [n]$ the n -tuples of free De Morgan polynomials in m variables, composition by substitution. This is the cube category with connections, reversals, degeneracies and diagonals, the maximal “classical” cube category in the taxonomy of [2, 3]. Our main theorem (Theorem 7.2) states:

Over a context of generic cells $q_1, \dots, q_s$ of arities $n_1, \dots, n_s$, the pure-reparametrization cubes modulo definitional equality form, naturally in $[m]$, the wedge of pointed sets

$$\bigvee_{t=1}^s \square_{\text{DM}}([m], [n_t]) / \partial,$$

where $\square_{\text{DM}}(-, [n]) / \partial$ collapses to the basepoint exactly the morphisms that factor through a face of $[n]$.

That is: each generic cell spans a *De Morgan sphere* — the boundary-collapsed representable — the spheres of distinct cells meet only at the common basepoint (the constant cube), and nothing else is identified. The action realizing the isomorphism is *strictly* functorial: the identity tuple acts as the identity and composition acts as substitution, both definitionally (machine-checked).

Three consequences delimit the strict/weak boundary exactly:

- (1) *Transposition is not strict*: for a generic 2-cell q , the squares $\langle i \rangle \langle j \rangle q(i)(j)$ and $\langle i \rangle \langle j \rangle q(j)(i)$ are distinct morphisms of $\square_{\text{DM}}$, hence not definitionally equal. This is the precise definitional shadow of the Eckmann–Hilton argument being weak: any exchange of coordinates requires composition structure, which lives outside the reparametrization layer.
- (2) *No excluded middle in any dimension*: $i \wedge \neg i$ is not an endpoint of the free De Morgan algebra, so the square $\langle i \rangle \langle j \rangle q(i \wedge \neg i)(j)$ — all four of whose *edges* are degenerate — is not definitionally constant. The free De Morgan algebra’s failure of Booleanness is definitional in every dimension.
- (3) *Cells do not interact*: no definitional equation relates cubes built from distinct generic cells, beyond the shared basepoint. Whiskering, horizontal and vertical composition, and the interchange law therefore live in the weak layer *by theorem*, not merely by inspection of the rules.

Honest positioning. The positive half of the characterization — that the De Morgan laws hold definitionally on reparametrized cells — is by design of the theory: the interval *is* a free De Morgan algebra and the kernel decides its equality by normal forms [1]. The content of this paper is the *converse* and its structure: nothing else is identified (faithfulness), what collapses is exactly face-factoring (the sphere structure), the collapse point is shared across cells (the wedge), and the action is strictly functorial. The one-dimensional case of Theorem 5.2 — a single generic path, $n = 1$ — appeared as a self-contained appendix of the companion paper [9]; the present paper strictly generalizes it (arbitrary arities, higher and multiple cells, the sphere and wedge structure, the strict functoriality of the action) and is written to be read independently. Beyond that special case we know of no prior statement or proof of this characterization. It does not follow from published normalization theorems: the normalization result of Sterling–Angiuli [8] concerns *Cartesian* cubical type theory, whose cube category has no connections or reversals, and open-term normalization for the De Morgan calculus is not in the literature. Our proofs are elementary given the right lemmas — they analyze the normalization-by-evaluation conversion of a concrete kernel — and every statement is, by the metatheory of the companion paper [9], a decidable and assumption-free fact about a minimal fragment of the calculus. Related coherence questions. The theorem is a coherence result of “every diagram commutes iff it commutes in the free structure” type [6]: a diagram of reparametrizations of a generic cell commutes definitionally iff it commutes in the free De Morgan algebra — and, complementing it, no diagram involving composition commutes definitionally unless trivially. The necessity of the weak layer is not an artifact of our kernel: by the no-go theorem of [9], forcing composition-layer equations (such as constant-tube collapse for `hcomp`) into the definitional equality of a cubical theory derives regularity principles that fail in the standard models. The present paper thus pins the strict/weak boundary from both sides: the strict side is exactly a wedge of De Morgan spheres, and crossing the boundary is not a matter of better engineering but is obstructed in principle.

Outline. Section 2 fixes the De Morgan cube category. Section 3 fixes the fragment of cubical type theory and its algorithmic equality, and states the standing metatheoretic facts imported from [9]. Section 4 defines the action and proves its strict functoriality. Section 5 proves the single-cell characterization (C_n), Section 6 the multi-cell wedge theorem, and Section 7 assembles the main theorem. Section 8 records the weak complement. Section 9 describes the machine verification.

2. THE DE MORGAN CUBE CATEGORY

Definition 2.1. $\text{DM}(i_1, \dots, i_m)$ is the free De Morgan algebra on generators $i_1, \dots, i_m$: the free bounded distributive lattice with an involution $\neg$ satisfying the De Morgan laws ($\neg(x \wedge y) = \neg x \vee \neg y$, $\neg 0 = 1$), and *without*

the laws $x \wedge \neg x = 0$, $x \vee \neg x = 1$. Its elements have canonical normal forms: antichains of clauses over the literals $i_k, \neg i_k$ (disjunctive normal form, reduced under absorption) [1]. We write 0, 1 for the bounds and call them the *endpoints*.

Definition 2.2 (The De Morgan cube category). $\square_{\text{DM}}$ has objects $[m]$ for $m \geq 0$ and morphisms

$$\square_{\text{DM}}([m], [n]) := \text{DM}(i_1, \dots, i_m)^n,$$

an n -tuple $\vec{\varphi} = (\varphi_1, \dots, \varphi_n)$ regarded as a substitution; composition is substitution of tuples into variables, and the identity on $[n]$ is the tuple of generators $(i_1, \dots, i_n)$.

$\square_{\text{DM}}$ is the classical cube category with faces, degeneracies, symmetries, diagonals, connections of both kinds, and reversals — in the taxonomy of Buchholtz–Morehouse [3] it is the cube category of the “De Morgan” variety; the study of cubical sites with connections goes back to Grandis–Mauri [2]. Two classes of morphisms play a role below:

Definition 2.3. The k -th face at $\varepsilon \in \{0, 1\}$ is $\delta_k^\varepsilon = (i_1, \dots, i_{k-1}, \varepsilon, i_k, \dots, i_{n-1}) : [n-1] \rightarrow [n]$. A morphism $\vec{\varphi} : [m] \rightarrow [n]$ is *face-factoring* if it factors through some δ_k^ε ; equivalently, iff some coordinate φ_k is an endpoint of the free algebra (i.e. its normal form is the empty or the full antichain).

Definition 2.4 (De Morgan spheres). $\square_{\text{DM}}(-, [n])/\partial$ is the pointed presheaf obtained from the representable $\square_{\text{DM}}(-, [n])$ by identifying all face-factoring morphisms to a single basepoint.

The subtlety that drives several results below is that face-factoring is a normal-form condition, not a pointwise one: $i \wedge \neg i$ takes the value 0 at both endpoints of i , yet it is *not* the endpoint 0 of the free algebra, so $(i \wedge \neg i, j)$ is not face-factoring.

3. THE CALCULUS, GENERIC CELLS, AND IMPORTED METATHEORY

3.1. The fragment. We work in a CCHM-style cubical type theory [1] with a De Morgan interval; everything below lives in the *minimal fragment*: Π , Σ , path types **Path**, and a base type, with transport and homogeneous composition — no universes, no glueing, no higher inductive types. In fact the terms of this paper use only path types over a base type: the fragment of *reparametrization*,

$$p \mapsto \langle i_1 \rangle \cdots \langle i_m \rangle p(\varphi_1) \cdots (\varphi_n), \quad \varphi_k \in \text{DM}(i_1, \dots, i_m),$$

where $p(\varphi)$ denotes path application and $\langle i \rangle$ path abstraction. Kan operations (**transp**, **hcomp**) are *excluded* from the layer under study; they are what the weak layer is made of (Section 8).

Definition 3.1 (Generic cells). Over a base context $A : \mathcal{U}$, $a : A$, the n -fold loop type is $\text{Loop}^1 := \text{Path } A a a$ and $\text{Loop}^{k+1} := \text{Path } (\text{Loop}^k) \text{ refl}_k \text{ refl}_k$,

where refl_k is the k -fold reflexivity tower. A *generic n -cell* is a context variable $q : \text{Loop}^n$. For $\vec{\varphi} \in \square_{\text{DM}}([m], [n])$ we write

$$A_q(\vec{\varphi}) := \langle i_1 \rangle \cdots \langle i_m \rangle q(\varphi_1) \cdots (\varphi_n),$$

an m -cube over the context.

Cells with fully degenerate boundary (loops) are the right generic objects for a *pointed* characterization; see Remark 7.3 for the expected unpointed refinement.

3.2. Algorithmic equality, and what we import. Definitional equality is taken to be the algorithmic equality $\equiv$ of a concrete normalization-by-evaluation kernel: terms are evaluated to weak-head values (path abstractions become closures; applications of variables become *neutral spines*), and conversion compares values type-directly, opening binders at fresh generic dimensions and comparing interval arguments of neutral path applications by the canonical antichain normal form of Definition 2.1. The kernel is the one of the companion paper [9], whose artifact this paper’s probes extend; three imported facts are standing hypotheses:

- (K1) *Endpoint collapse is normal-form exact.* Path application of a *value* at an interval argument r collapses to the stored endpoint iff r is an endpoint of the free De Morgan algebra (the kernel decides $r = 0, 1$ by the antichain normal form, not syntactically).
- (K2) *Conversion of neutrals is spine-wise.* Two neutral spines are convertible iff they have the same head variable, the same length, convertible stored endpoint annotations, and, at each path-application stage, interval arguments equal in the free De Morgan algebra.
- (K3) *Decidability without assumptions.* On the minimal fragment, evaluation and conversion terminate on well-typed input and the algorithmic equality is decidable; no base-theory assumption is involved. (This is the endpoint-blind termination theorem of [9].)

(K1) and (K2) are facts about two specific functions of the kernel, verified there; (K3) is a theorem of [9]. Every positive and negative instance claimed below is additionally machine-checked (Section 9); by (K3) each such check is a decision, not a test.

Remark 3.2 (Robustness). The reparametrization layer involves no transport, so the results are insensitive to the transport-layer design choices studied in [9] (in particular to its evaluation-time constancy rule): they hold verbatim for a standard CCHM-style conversion. We state them for our kernel because there they are *unconditional* decidable facts by (K3).

4. THE ACTION AND ITS STRICT FUNCTORIALITY

Reparametrization defines, for each generic n -cell q and each m , a function

$$A_q : \square_{\text{DM}}([m], [n]) \longrightarrow \{m\text{-cubes over } \Gamma\}, \quad \vec{\varphi} \mapsto A_q(\vec{\varphi}).$$

Lemma 4.1 (Unit law, definitionally). $A_q(\text{id}_{[n]}) \equiv q$.

Proof. $A_q(\text{id}) = \langle i_1 \rangle \cdots \langle i_n \rangle q(i_1) \cdots (i_n)$, which is $\equiv q$ by n applications of the η -rule for path types (conversion is type-directed: it opens both sides at generic dimensions and compares the resulting applications, which are literally equal spines). Machine probe: **sqEtaD**. $\square$

Lemma 4.2 (Composition law, definitionally). For $\vec{\varphi} : [m] \rightarrow [n]$ and $\vec{\psi} : [l] \rightarrow [m]$,

$$(A_q(\vec{\varphi})) \text{ reparametrized along } \vec{\psi} \equiv A_q(\vec{\varphi} \circ \vec{\psi}),$$

where $\vec{\varphi} \circ \vec{\psi}$ is substitution in $\square_{\text{DM}}$.

Proof. Path application of a path abstraction is β -reduction, which the evaluator implements as substitution of the interval argument into the body; iterating, the left-hand side evaluates to the right-hand side's evaluation. Machine probe: **sqCloneCompD** ($((\langle u \rangle \langle v \rangle q(u)(v))(i \vee j)(i \wedge j) \equiv q(i \vee j)(i \wedge j))$). $\square$

Thus A_q is *strictly* functorial — the action of $\square_{\text{DM}}$ on cubes is by definitional equality, with no coherence cells needed. This is the precise sense in which the connection layer of cubical type theory is “strict algebra”: the entire free De Morgan theory acts on cells on the nose.

5. THE SINGLE-CELL CHARACTERIZATION

Fix a generic n -cell q . The following computation is the engine of everything.

Lemma 5.1 (Value computation). *Open the m binders of $A_q(\vec{\varphi})$ at generic dimensions. The body $q(\varphi_1) \cdots (\varphi_n)$ evaluates as follows.*

- (1) *If some φ_k is an endpoint of the free algebra, the value is the base point a : the first such coordinate collapses the spine — by (K1), either the application of q itself (yielding a reflexivity tower, through which the remaining applications compute), or the application of an already-neutral spine (yielding its stored endpoint annotation, again a tower) — and towers evaluate through the remaining arguments to a .*
- (2) *Otherwise the value is the full neutral spine $q(\varphi_1) \cdots (\varphi_n)$ with all n interval arguments retained.*

In particular the value is a iff $\vec{\varphi}$ is face-factoring (Definition 2.3).

Proof. By (K1), the collapse at each stage happens iff the argument is a free-algebra endpoint; the case analysis is exhaustive. Machine probes: **sq2DegFstD** (collapse at the first coordinate), **sq2DegSndD** (collapse of a neutral spine at the second), **sq2DegCrossD** (all face-factoring squares are equal). $\square$

Theorem 5.2 (C_n : the sphere of a single cell). *For $\vec{\varphi}, \vec{\psi} \in \square_{\text{DM}}([m], [n])$ (with boundaries agreeing so that the comparison is well-typed):*

$$A_q(\vec{\varphi}) \equiv A_q(\vec{\psi}) \iff \begin{cases} \text{both are face-factoring, or} \\ \text{neither is, and } \vec{\varphi} = \vec{\psi} \text{ in } \square_{\text{DM}}([m], [n]). \end{cases}$$

Hence A_q descends to an injection $\square_{\text{DM}}([m], [n])/\partial \hookrightarrow \{m\text{-cubes}\}/\equiv$ of pointed sets, the basepoint being the constant cube.

Proof. Conversion opens the m binders at generic dimensions and compares the bodies of Lemma 5.1.

($\Leftarrow$) If both tuples are face-factoring, both bodies are the same variable a . If neither is and the tuples are equal, the spines agree stage-wise: same head, same stored endpoint annotations (the same towers), and interval arguments equal in the free algebra, which the antichain normal form decides (K2).

($\Rightarrow$) Suppose conversion accepts. If both bodies are a , both tuples are face-factoring by Lemma 5.1. If both are spines, (K2) forces stage-wise equality of the interval arguments in the free algebra, i.e. $\vec{\varphi} = \vec{\psi}$. A variable cannot be convertible with a spine, so the mixed case is impossible. $\square$

Corollary 5.3 (Transposition is not strict). *For a generic 2-cell, $\langle i \rangle \langle j \rangle q(j)(i) \neq \langle i \rangle \langle j \rangle q(i)(j)$.*

Proof. $(j, i) \neq (i, j)$ in $\square_{\text{DM}}([2], [2])$, and neither is face-factoring. Machine probe with a well-typedness control. $\square$

Corollary 5.4 (No excluded middle in dimension 2). *$\langle i \rangle \langle j \rangle q(i \wedge \neg i)(j) \neq \langle i \rangle \langle j \rangle a$, although all four edges of the left-hand square are degenerate.*

Proof. $i \wedge \neg i$ is not an endpoint of the free algebra (its antichain normal form is the singleton clause $\{i, \neg i\}$), so $(i \wedge \neg i, j)$ is not face-factoring; apply Theorem 5.2 against the (constant) face-factoring class. Machine probes `sq2LEMCtrlD` and its companion guard. $\square$

Remark 5.5. Corollary 5.4 is the higher-dimensional face of a familiar dimension-1 fact ($\langle i \rangle p(i \wedge \neg i) \neq \text{refl}$): the free De Morgan algebra is not Boolean, and the definitional equality sees this in every dimension — a square all of whose edges are constant need not be definitionally constant.

6. MULTIPLE CELLS: THE WEDGE

Observation 6.1 (Syntactic exhaustiveness). Over a context with generic cells $q_1 : \text{Loop}^{n_1}, \dots, q_s : \text{Loop}^{n_s}$ (common base point a) and no Kan operations, every definable body of an m -cube evaluates either to a or to an application spine of exactly one q_t : the syntax has no interval case-split, so a single body term cannot mix two cells.

Theorem 6.2 (Wedge theorem). *For cells q_t, q_u and tuples $\vec{\varphi}, \vec{\psi}$:*

$$A_{q_t}(\vec{\varphi}) \equiv A_{q_u}(\vec{\psi}) \iff \begin{cases} \text{both tuples are face-factoring (any } t, u), \text{ or} \\ t = u, \text{ neither is, and } \vec{\varphi} = \vec{\psi}. \end{cases}$$

Proof. Lemma 5.1 applies to each cell separately. Face-factoring bodies are the *same* variable a regardless of the cell — the shared basepoint. Non-collapsed spines are compared by (K2), which separates distinct head variables ($t \neq u$), distinct spine lengths ($n_t \neq n_u$), and distinct tuples. Machine probes: **sqxDegD** (cross-cell collapse: $\langle i \rangle \langle j \rangle q(0)(j) \equiv \langle i \rangle \langle j \rangle r(i)(1)$), head separation, and spine-depth separation (a 1-cell square built with the nonzero formula $(i \wedge \neg i) \wedge (j \wedge \neg j)$ — all four edges degenerate, interior nonzero — against a 2-cell square). $\square$

7. THE MAIN THEOREM

One more lemma extends the characterization from full applications to all cubes of higher loop types (partial applications).

Lemma 7.1 (Partial applications). *An m -cube valued in $\mathbf{Loop}^d$ definable in the layer is a partial application $\langle \vec{i} \rangle_{q_t}(\varphi_1) \cdots (\varphi_{n_t-d})$ (or a tower). Type-directed conversion at $\mathbf{Loop}^d$ opens d further binders at fresh generic dimensions; since a generic dimension is never a free-algebra endpoint, the comparison reduces to Theorem 6.2 for the extended tuples $(\varphi_1, \dots, \varphi_{n_t-d}, l_1, \dots, l_d)$, which are face-factoring iff the original partial tuples are.*

Proof. Conversion at path types is η -directed: it compares applications at a generic dimension. Iterating d times appends d distinct generic dimensions to the spine, none of them an endpoint. Machine probes: the diagonal $[1] \rightarrow [2]$ (**sqDiagCtrlD**) and the diagonal/anti-diagonal separation. $\square$

Theorem 7.2 (Main theorem: a wedge of De Morgan spheres). *Over a context of generic cells $q_1, \dots, q_s$ of arities $n_1, \dots, n_s$ with common base point, the sets of pure-reparametrization m -cubes modulo algorithmic equality assemble, naturally in $[m] \in \square_{\mathbf{DM}}^{\text{op}}$, into the wedge of pointed presheaves*

$$\{\text{cubes}\} / \equiv \cong \bigvee_{t=1}^s \square_{\mathbf{DM}}(-, [n_t]) / \partial.$$

The isomorphism is induced by the actions A_{q_t} ; naturality is Lemmas 4.1 and 4.2 (the actions are strictly functorial); injectivity is Theorems 5.2 and 6.2; surjectivity is Observation 6.1 with Lemma 7.1. All of this is decidable and assumption-free by (K3).

Remark 7.3 (Boundaries; the unpointed refinement). Our generic cells are loops: their boundaries are reflexivity towers, and it is exactly this degeneracy that produces the $/\partial$ collapse (Lemma 5.1 computes the collapse *through* the towers). For a generic cell with *generic boundary* — a context supplying fresh lower-dimensional cells for each face — the same analysis should yield the full representable $\square_{\mathbf{DM}}(-, [n])$ with no collapse, and gluing the two computations along boundary inclusion should give a cofibration of presheaves. We leave this unpointed refinement, and the general position of cell contexts among $\square_{\mathbf{DM}}$ -computads, to future work.

8. THE WEAK COMPLEMENT

Theorem 7.2 is exhaustive for the layer without Kan operations; everything else about the ω -groupoid structure of types — composition, inverses’ cancellation, units of composition, whiskering, interchange, Eckmann–Hilton — involves `hcomp` and is *weak*. Two points deserve emphasis.

First, weakness is now a *theorem* rather than an observation: by Theorem 6.2, no definitional equation relates cubes built from distinct cells, so no candidate “definitional composition” of generic cells exists in the calculus at all; and by Corollary 5.3, even the one-cell exchange of coordinates fails definitionally.

Second, the boundary is not an engineering accident. The companion paper [9] proves a no-go theorem: extending the definitional equality of a cubical theory with composition-layer collapse (e.g. equating a constant-tube `hcomp` with its base) derives regularity principles of the kind that fail in the standard cubical-sets models, and the failure arises already at path types. Combined with Theorem 7.2, the picture is complete: *the strict layer of generic cells is exactly a wedge of De Morgan spheres, and it cannot be enlarged*.

For calibration, the machine-checked strictness catalogue of [9] sits exactly on this boundary: on the strict side, all connection reparametrizations, η -laws, and transport-related equations up to and including a definitional `J` on `refl`; on the weak side, all `hcomp`-generated laws, beginning with `trans refl refl` $\neq$ `refl`.

9. MACHINE VERIFICATION

All positive and negative instances quoted in this paper are elaboration-time checks in a self-contained cubical kernel written in Lean 4 (no dependence on Lean’s definitional equality), the artifact of [9]: repository <https://github.com/r-shibusawa/cubical-strict-j>. The probes of this paper (`LibSquares.lean`, together with the relevant rows of `LibStrictness.lean` and the supporting proof document `docs/ReparamCoherence2.md`) are included from release `v1.2.0`, archived at DOI 10.5281/zenodo.21638543; the companion paper’s own snapshot is release `v1.1.0`, DOI 10.5281/zenodo.21637898. A successful build replays every check. The probes are decision procedures by (K3): each catalogue row is a decided fact about the algorithmic equality, not a test that might be flaky.

The division of labor is the usual one for this artifact: the kernel supplies machine-checked *instances* (both identifications and separations); the theorems of this paper are pen-and-paper results about the conversion algorithm, whose load-bearing clauses (K1), (K2) are small, inspectable functions.

ACKNOWLEDGEMENTS

The metatheoretic platform (the kernel, the strictness catalogue, and the decidability theorem for the minimal fragment) was developed in the

companion work [9], whose anonymous review rounds indirectly shaped the present formulation.

REFERENCES

- [1] C. Cohen, T. Coquand, S. Huber, A. Mörtberg. Cubical type theory: a constructive interpretation of the univalence axiom. *TYPES 2015*, LIPIcs 69, 2018.
- [2] M. Grandis, L. Mauri. Cubical sets and their site. *Theory and Applications of Categories* 11 (2003), 185–211.
- [3] U. Buchholtz, E. Morehouse. Varieties of cubical sets. *RAMiCS 2017*, LNCS 10226; arXiv:1701.08189.
- [4] P. L. Lumsdaine. Weak ω -categories from intensional type theory. *Logical Methods in Computer Science* 6(3), 2010.
- [5] B. van den Berg, R. Garner. Types are weak ω -groupoids. *Proceedings of the London Mathematical Society* 102(2), 2011, 370–394.
- [6] S. Mac Lane. Natural associativity and commutativity. *Rice University Studies* 49(4), 1963, 28–46.
- [7] A. Vezzosi, A. Mörtberg, A. Abel. Cubical Agda: a dependently typed programming language with univalence and higher inductive types. *Journal of Functional Programming* 31, 2021.
- [8] J. Sterling, C. Angiuli. Normalization for cubical type theory. *LICS 2021*.
- [9] R. Shibusawa. Evaluation-time constancy inference for transport in cubical type theory. Preprint, 2026. Artifact: <https://github.com/r-shibusawa/cubical-strict-j>, archived at DOI 10.5281/zenodo.21637898.

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